

■ AgriFlow: Smart Agriculture & Farm Intelligence

Full Pitch Dossier, Technical Architecture, System Stack & Cross-Question Defense

1. Executive Summary & Value Proposition

AgriFlow is an end-to-end Precision Agriculture, Real-Time IoT Telemetry, and Farm Financial Operating System designed specifically for smallholder and commercial farmers. By uniting **Multimodal Vision AI** for plant pathology, **Real-Time Physical IoT Sensors** via Supabase, **Hyper-Local Peer Mandi Trade Transparency**, and an **Automated Profit & Loss Farm Ledger** into an intuitive multilingual mobile interface (Marathi, Hindi, English), AgriFlow empowers farmers to lower chemical input costs by 25–35%, prevent devastating crop diseases, optimize irrigation cycles, and maximize market selling rates.

2. The Problem: Systemic Gaps in Grassroots Agriculture

- **Delayed Disease Diagnosis & Toxic Overdosing:** Farmers rely on input retailers who push expensive chemical cocktails, killing beneficial soil biology and escalating debts.
- **Blind Irrigation & Water Stress:** Lack of affordable ground moisture sensors leads to either fungal root rot from waterlogging or yield-stunting drought stress.
- **Mandi Price Asymmetry & Trader Exploitation:** Farmers lack real-time visibility into what neighboring peers sold identical crops for at local APMC mandis on that very morning.
- **Absence of Financial Recordkeeping:** Over 90% of smallholders maintain no formal income/expense ledgers, leaving them unable to assess crop profitability or qualify for structured credit.
- **Language & Usability Barrier:** Most agritech applications feature complex, text-dense English interfaces unsuitable for rural farmers.

3. The AgriFlow Solution: 6 Core Pillars

Pillar	Functionality & Impact
1. AI Vision Leaf Scanner	Instant computer-vision disease diagnosis with confidence scoring. Delivers dual treatment paths: Chemical (exact brand name e.g. Bayer Saaf + pump dosage) and Organic (Neem oil, Trichoderma viride) + 1-tap ledger expense integration.
2. Live IoT Soil Telemetry	Direct integration with field ESP32 nodes streaming Soil Moisture, Ambient Temperature, and Air Humidity to Supabase in real time with zero-cache delay and a 5-entry rolling FIFO history table.
3. 5-Day Plant Lifecycle Engine	Structured developmental tracking across phenological stages (Sowing → Vegetative → Flowering → Harvest) with strict 5-day photo validation and proactive preventative care advisory.
4. Real Peer Mandi Trades	Hyper-local trade sharing where farmers in nearby districts (e.g. Nagpur, Ramtek, Saoner) view verified sales recorded by actual peers with transparent rate slips (■/kg and ■/quintal).
5. Farm Ledger & Net Savings	Automated income and expense tracking categorized into Seeds, Fertilizers, Labor, Machinery, and Pesticides with dynamic Net Farm Savings (■■■■■■ ■■■■) computation.
6. Native Multilingual Engine	100% Devanagari regional support across English, Marathi (■■■■■■), and Hindi (■■■■■■). All AI advisory, pathology reports, mandi trades, and financial analytics dynamically adapt.

4. Comprehensive Technical Architecture & Tech Stack

AgriFlow is engineered using a robust, decoupled microservices architecture designed for high availability, low mobile data consumption, and sub-second cold starts.

Layer / Subsystem	Technologies Used	Implementation Details & Role
Frontend Mobile App	React Native Expo SDK 57 Expo Router Lucide React Native AsyncStorage	Cross-platform (Android & iOS) client. File-based routing, reactive State Contexts (Auth, Language, Crop), zero-flicker native animations, offline token storage, sub-second boot optimization.
Backend API Server	Python FastAPI Uvicorn ASGI Pydantic Models Bcrypt Hashing	High-performance asynchronous REST API running on Port 8001. Handles secure authentication, diagnostic request routing, financial analytics, and proxy telemetry fallback.
Relational Database	SQLite 3 (agriflow.db)	Embedded relational database storing structured tables: <i>farmers</i> , <i>farmer_crops</i> , <i>scans</i> , <i>finances</i> , <i>community_trades</i> , <i>admin_users</i> with foreign keys and index optimization.
Cloud IoT & Telemetry	Supabase PostgreSQL REST Zero-Cache API	Direct cloud ingestion layer for hardware sensor payloads. Direct REST API queries with dynamic 10-second polling and rolling 5-entry FIFO history persistence.
AI Vision & LLM Engine	Groq Cloud Vision Llama 3.2 Vision Qwen 2.5 Agronomy Rule Base	Sub-second multimodal leaf analysis for pathology detection. Strict system-prompt sandboxing against ICAR/CIBRC dosage standards to eliminate hallucinations.
Physical IoT Hardware	ESP32 Microcontroller Capacitive Soil Probe DHT22 / DHT11 Sensor Solar Li-Ion Node	Low-power Wi-Fi/GSM IoT node measuring volumetric soil moisture (12-bit ADC mapping) and ambient temperature/humidity. Automated 3-minute transmission cycles.

5. Business Model & Commercialization Strategy

- **Freemium Farmer Utility (B2C):** Free basic leaf scanning, community mandi pricing, and offline farm diary to drive viral adoption across agricultural belts.
- **Agri-Input Affiliate & Marketplace Commissions:** Partnering with certified agrochemical and bio-fertilizer brands (Bayer, Syngenta, IFFCO). AgriFlow earns a **3–7% fulfillment commission** when farmers purchase prescribed remedies.
- **Hardware Node Sales & Subscriptions:** Sale of pre-calibrated ESP32 IoT sensor kits at a 30% gross margin with optional premium predictive irrigation alerting tiers.
- **B2B Enterprise & FPO SaaS Licensing:** Web management portal for Farmer Producer Organizations (FPOs), micro-finance institutions, and corporate contract farming organizations to track yield health and creditworthiness across clusters.

6. Ready-to-Use Pitch Scripts

■ ■ 30-Second Elevator Pitch:

"Every season, Indian farmers lose up to 35% of their yield to preventable crop diseases, blind irrigation, and unfair mandi pricing. AgriFlow is the all-in-one smart farming companion. By combining instant AI leaf disease diagnosis, live ESP32 soil sensor telemetry, hyper-local peer mandi trade sharing, and automated financial bookkeeping in native languages like Marathi and Hindi, AgriFlow empowers farmers to cut input costs by 25% and maximize harvest profits."

7. Comprehensive Cross-Questioning Defense & Q&A;

Prepare for jury, investor, and academic cross-examination with these structured responses:

Q1: What is your core competitive moat against established apps like Plantix or AgroStar?

Plantix is strictly an isolated leaf scanner—it has no physical IoT telemetry, financial ledger, or peer mandi network. AgroStar and DeHaat are primarily input e-commerce companies incentivized to push their own chemical inventory. AgriFlow is an unbiased, full-stack operating system unifying Vision AI + Real-time Hardware Telemetry + Peer Mandi Rates + P&L; Accounting into one cohesive regional app.

Q2: How do you prevent AI hallucinations when prescribing chemical dosages?

AgriFlow uses a two-tier safety architecture. The multimodal model's output is strictly sandboxed against a curated ICAR/CIBRC agricultural knowledgebase. Dosages are bounded by standard 15L water pump spray benchmarks (e.g. 2g/L) and paired with both chemical brand names (Bayer Saaf) and safe organic biological remedies (Neem Oil, Trichoderma).

Q3: What happens if a farmer uploads a blurry photo or a non-crop object?

Our vision pipeline includes a pre-validation clarity filter. If the image lacks sufficient leaf morphology or lighting, the app immediately prompts the user to retake a clear leaf photograph in natural sunlight. For borderline scans, it flags the diagnosis as 'Under Review' and defaults to non-toxic broad-spectrum bio-fungicides.

Q4: How does the application function in rural areas with poor or zero network connectivity?

AgriFlow is engineered offline-first. User profiles, farm metadata, previously analyzed scan reports, and financial ledgers are persisted locally using SQLite and AsyncStorage. Sensor data queries employ a dual pipeline: direct Supabase REST endpoints with automatic backend proxy fallback.

Q5: Is an ESP32 sensor hardware node durable and affordable for smallholder farmers?

Commercial agricultural stations cost ₹25,000 to ₹60,000, making them inaccessible. Our node utilizes an ESP32 SoC paired with a corrosion-resistant capacitive soil moisture probe and DHT climate sensor, costing under ₹1,200 (\$14 USD). Operating on ultra-low-power deep sleep cycles, it can run for months on a small solar-backed lithium cell.

Q6: How do you verify that community mandi prices are genuine and not manipulated?

Community mandi trades require authenticated farmer logins with location tracking. The app cross-references reported commodity rates against APMC historical daily bands to detect anomalies. Furthermore, verified peer sales show farmer name, village, mandi name, and exact quantity sold.

Q7: Why enforce a 5-day photo interval in the Plant Lifecycle Tracker?

Crop vegetative morphology and disease incubation cycles take several days to manifest measurable changes. The 5-day interval prevents photo spamming, builds disciplined observational habits for the farmer, and aligns perfectly with preventative spray schedules.